Linear Discriminant Analysis (LDA)

Multiple Choice Questions & Solutions Manual

1. What is the primary objective of Linear Discriminant Analysis (LDA)?

A) Minimize reconstruction error

B) Maximize class separability CORRECT ANSWER

C) Minimize classification accuracy

D) Increase dimensionality

2. LDA is primarily used for:

A) Clustering

B) Regression

C) Dimensionality Reduction and Classification CORRECT ANSWER

D) Data Augmentation

3. Which matrices are central to the formulation of LDA?

A) Covariance and Correlation Matrices

B) Within-Class Scatter and Between-Class Scatter Matrices CORRECT ANSWER

C) Adjacency and Incidence Matrices

D) Kernel and Gram Matrices

4. The within-class scatter matrix represents:

A) Separation between class means

B) Variance of samples within each class CORRECT ANSWER

C) Distance between clusters

D) Prediction error

5. The between-class scatter matrix measures:

A) Variability within classes

B) Separation among class means CORRECT ANSWER

C) Noise variance

D) Reconstruction loss

6. LDA seeks projection directions that:

A) Minimize both within-class and between-class variance

B) Maximize within-class variance

C) Maximize between-class variance while minimizing within-class variance CORRECT ANSWER

D) Ignore class labels

7. For a dataset with C classes, the maximum number of LDA components is:

A) C

B) C + 1

C) C - 1 CORRECT ANSWER

D) Equal to the number of features

8. Which of the following is an assumption of classical LDA?

A) Classes have identical covariance matrices CORRECT ANSWER

B) Classes have different covariance matrices

C) Features are independent and binary

D) Data must be non-linear

9. LDA is a supervised technique because:

A) It uses unlabeled data

B) It requires class labels CORRECT ANSWER

C) It uses reinforcement signals

D) It performs clustering

10. In LDA, eigenvectors are obtained from:

A) Covariance Matrix

B) Correlation Matrix

C) $S_W^{-1} S_B$ CORRECT ANSWER

D) Gram Matrix

11. Compared to PCA, LDA:

A) Does not reduce dimensions

B) Ignores class labels

C) Uses class information CORRECT ANSWER

D) Requires more dimensions

12. Which criterion is maximized in Fisher's Linear Discriminant?

A) Reconstruction Error

B) Variance Ratio

C) Between-Class Scatter to Within-Class Scatter Ratio CORRECT ANSWER

D) Mean Squared Error

13. LDA assumes data from each class follows:

A) Uniform Distribution

B) Poisson Distribution

C) Gaussian Distribution CORRECT ANSWER

D) Exponential Distribution

14. If there are two classes, the maximum number of discriminant components is:

A) 0

B) 1 CORRECT ANSWER

C) 2

D) Number of features

15. Which statement best differentiates PCA and LDA?

A) PCA is supervised; LDA is unsupervised

B) PCA maximizes variance; LDA maximizes class separability CORRECT ANSWER

C) PCA requires labels; LDA does not

D) Both optimize the same objective

16. What happens if class labels are randomly shuffled before applying LDA?

A) Performance generally degrades CORRECT ANSWER

B) Performance improves

C) No effect

D) Number of features increases

17. Which matrix must generally be invertible in standard LDA?

A) Between-Class Scatter Matrix

B) Within-Class Scatter Matrix CORRECT ANSWER

C) Correlation Matrix

D) Distance Matrix

18. LDA projects data onto:

A) Directions of maximum variance

B) Random directions

C) Directions maximizing discrimination among classes CORRECT ANSWER

D) Kernel feature space only

19. In pattern recognition, LDA is often applied:

A) Before classification CORRECT ANSWER

B) After classification

C) Only during testing

D) Only for clustering

20. Which of the following can be a limitation of LDA?

A) Requires labeled data

B) Assumes linear boundaries

C) Sensitive to covariance estimation

D) All of the above CORRECT ANSWER

21. LDA can be viewed as:

A) Feature Extraction Technique

B) Data Compression Technique

C) Classification Technique

D) All of the above CORRECT ANSWER

22. The rank of the between-class scatter matrix is at most:

A) Number of Features

B) Number of Samples

C) Number of Classes Minus One CORRECT ANSWER

D) Twice the Number of Classes

23. Which application commonly uses LDA?

A) Face Recognition

B) Speech Recognition

C) Medical Diagnosis

D) All of the above CORRECT ANSWER

24. In Fisher's criterion, a good projection should:

A) Bring class means closer together

B) Increase overlap among classes

C) Separate class means and reduce class spread CORRECT ANSWER

D) Ignore variance

25. LDA is most effective when:

A) Classes are well separated CORRECT ANSWER

B) Data is completely random

C) No labels are available

D) Features are constant

26. Which of the following affects LDA performance?

A) Class imbalance

B) Outliers

C) Small sample size

D) All of the above CORRECT ANSWER

27. The generalized eigenvalue problem in LDA is solved to obtain:

A) Cluster Centroids

B) Principal Components

C) Discriminant Directions CORRECT ANSWER

D) Decision Trees

28. LDA is likely to perform poorly when:

A) Classes are linearly separable

B) Covariance assumptions are heavily violated CORRECT ANSWER

C) Labels are accurate

D) Features are informative

29. Which quantity is minimized in the denominator of Fisher's criterion?

A) Between-Class Scatter

B) Within-Class Scatter CORRECT ANSWER

C) Total Variance

D) Correlation

30. In multiclass classification, LDA creates:

A) One discriminant axis only

B) Multiple discriminant axes up to C-1 CORRECT ANSWER

C) One axis per feature

D) No projection axes

31. LDA is particularly useful when:

A) Number of features is very small

B) Feature reduction is needed before classification CORRECT ANSWER

C) Data contains no labels

D) Only clustering is required

32. Which assumption is shared by many LDA formulations?

A) Equal class priors

B) Equal covariance matrices across classes CORRECT ANSWER

C) Independent samples only

D) Nonlinear decision surfaces

33. The output of LDA is typically:

A) Cluster Assignments

B) Reduced-Dimensional Feature Vectors CORRECT ANSWER

C) Regression Coefficients

D) Association Rules

34. Why is LDA often preferred over PCA for classification tasks?

A) It ignores variance

B) It uses label information CORRECT ANSWER

C) It requires fewer samples

D) It is unsupervised

35. If there are 5 classes in a dataset, the maximum number of discriminant vectors produced by LDA is:

A) 3

B) 4 CORRECT ANSWER

C) 5

D) 6

36. Which type of learning does LDA belong to?

A) Supervised Learning CORRECT ANSWER

B) Unsupervised Learning

C) Reinforcement Learning

D) Self-Supervised Learning

37. What role do class means play in LDA?

A) No role

B) Used to compute between-class scatter CORRECT ANSWER

C) Used only for normalization

D) Used only during testing

38. Which of the following is true about LDA decision boundaries?

A) Always nonlinear

B) Typically linear CORRECT ANSWER

C) Always quadratic

D) Always circular

39. LDA can suffer from the "small sample size" problem when:

A) Number of samples is much smaller than number of features CORRECT ANSWER

B) Number of features is zero

C) Classes are balanced

D) Data is normalized

40. Which statement about LDA is correct?

A) It ignores class membership

B) It maximizes variance without labels

C) It seeks projections that improve class discrimination CORRECT ANSWER

D) It is purely a clustering algorithm

Quick Answer Key Table

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	11	C	21	D	31	B
2	C	12	C	22	C	32	B
3	B	13	C	23	D	33	B
4	B	14	B	24	C	34	B
5	B	15	B	25	A	35	B
6	C	16	A	26	D	36	A
7	C	17	B	27	C	37	B
8	A	18	C	28	B	38	B
9	B	19	A	29	B	39	A
10	C	20	D	30	B	40	C